

Project Overview

This project focuses on exploratory data analysis (EDA) to understand the structure, quality, and patterns present in a real-world dataset.

The analysis will include data inspection, data-quality checks, descriptive statistics, missing-value analysis, duplicate detection, outlier identification, and exploration of relationship between important variables.

The goal is to understand the dataset before moving toward deeper analysis and visualization.

✓ 1. Questions to Explore

Before analyzing the dataset, the following questions will be investigated:

1. What are the main characteristics of the customers and their shopping behavior?
2. Which product categories are the most frequently purchased?
3. How does the purchase amount vary across different product categories?
4. Is there any relationship between customer age and purchase amount?
5. Do customer ratings differ across product categories?
6. Are certain payment methods or shopping preferences more common among customers?
7. Are there any unusual values, missing values, duplicate records, or other data-quality issues that could affect the analysis?
8. Can statistical analysis provide evidence for any relationship between important variables in the dataset?

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("/shopping_trends (1).csv")
```

```
df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	
1	2	19	Male	Sweater	Clothing	64	Maine	L	M
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	M
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	M
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turc

Next steps:

Generate code with df

New interactive sheet

```
df.shape
```

```
(3900, 19)
```

```
df.columns
```

```
Index(['Customer ID', 'Age', 'Gender', 'Item Purchased', 'Category',  
      'Purchase Amount (USD)', 'Location', 'Size', 'Color', 'Season',  
      'Review Rating', 'Subscription Status', 'Payment Method',  
      'Shipping Type', 'Discount Applied', 'Promo Code Used',  
      'Previous Purchases', 'Preferred Payment Method',  
      'Frequency of Purchases'],  
      dtype='object')
```

```
df.info
```

pandas.core.frame.DataFrame.info

```
def info(verbose: bool | None=None, buf: WriteBuffer[str] | None=None,  
max_cols: int | None=None, memory_usage: bool | str | None=None,  
show_counts: bool | None=None) -> None
```

</usr/local/lib/python3.13/dist-packages/pandas/core/frame.py>

Print a concise summary of a DataFrame.

This method prints information about a DataFrame including the index dtype and columns, non-null values and memory usage.

```
df.dtypes
```

	0
Customer ID	int64
Age	int64
Gender	object
Item Purchased	object
Category	object
Purchase Amount (USD)	int64
Location	object
Size	object
Color	object
Season	object
Review Rating	float64
Subscription Status	object
Payment Method	object
Shipping Type	object
Discount Applied	object
Promo Code Used	object
Previous Purchases	int64
Preferred Payment Method	object
Frequency of Purchases	object

dtype: object

✓ Variable Classification

The dataset contains the following types of variables:

Numerical Variables

- Customer ID
- Age
- Purchase Amount (USD)
- Review Rating

- Previous Purchases

Categorical Variables

- Gender
- Item Purchased
- Category
- Location
- Size
- Color
- Season
- Subscription Status
- Payment Method
- Shipping Type
- Discount Applied
- Promo Code Used
- Preferred Payment Method
- Frequency of Purchases

Customer ID is treated as an identifier rather than a variable for statistical analysis.

```
missing_values = df.isnull().sum()
print(missing_values)
```

```
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    0
Subscription Status  0
Payment Method   0
Shipping Type    0
Discount Applied  0
Promo Code Used  0
Previous Purchases  0
Preferred Payment Method  0
Frequency of Purchases  0
dtype: int64
```

```
missing_percentage = (df.isnull().sum() / len(df)) * 100
print(missing_percentage)
```

```
Customer ID      0.0
Age              0.0
Gender           0.0
Item Purchased   0.0
Category         0.0
Purchase Amount (USD) 0.0
Location         0.0
Size            0.0
Color           0.0
Season          0.0
Review Rating    0.0
Subscription Status 0.0
Payment Method   0.0
Shipping Type    0.0
Discount Applied 0.0
Promo Code Used  0.0
Previous Purchases 0.0
Preferred Payment Method 0.0
Frequency of Purchases 0.0
dtype: float64
```

✓ Missing Value Finding

The missing-value analysis shows that there are no missing values in any column of the dataset. Therefore, no missing-value treatment is required at this stage.

```
duplicate_count = df.duplicated().sum()
print(duplicate_count)
```

```
0
```

✓ Duplicate Value Finding

The duplicate-value check identified 0 duplicate rows in the dataset. Therefore, no duplicate records need to be removed at this stage.

```
df.describe()
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3900.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.749949	25.351538
std	1125.977353	15.207589	23.685392	0.716223	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.700000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

Descriptive Statistics Findings The dataset contains 3,900 customer records. The average customer age is approximately 44 years, while the average purchase amount is about 59.76 USD. The average review rating is approximately 3.75, indicating generally moderate-to-positive ratings. Previous purchases range from 1 to 50, with an average of approximately 25 purchases. The numerical variables will be explored further to identify distributions, patterns, and potential anomalies.

```
df.select_dtypes(include=['object']).nunique()
```

	0
Gender	2
Item Purchased	25
Category	4
Location	50
Size	4
Color	25
Season	4
Subscription Status	2
Payment Method	6
Shipping Type	6
Discount Applied	2
Promo Code Used	2
Preferred Payment Method	6
Frequency of Purchases	7

dtype: int64

Categorical Variable Analysis

The dataset contains several categorical variables with different numbers of unique values. Gender, subscription status, discount applied, and promo code used contain two categories, while variables such as item purchased, location, color, payment method, and purchase frequency contain multiple categories. These variables will be explored further to identify customer preferences, purchasing patterns, and differences between groups.

```
df.isnull().sum()
```

	0
Customer ID	0
Age	0
Gender	0
Item Purchased	0
Category	0
Purchase Amount (USD)	0
Location	0
Size	0
Color	0
Season	0
Review Rating	0
Subscription Status	0
Payment Method	0
Shipping Type	0
Discount Applied	0
Promo Code Used	0
Previous Purchases	0
Preferred Payment Method	0
Frequency of Purchases	0

dtype: int64

Missing Value Analysis

The missing-value analysis shows that all columns contain zero missing values. Therefore, the dataset does not require any missing-value imputation or removal at this stage. This indicates that the available records are complete with respect to the variables included in the dataset.

```
df.duplicated().sum()
```

```
np.int64(0)
```

Duplicate Record Analysis

The dataset contains 0 duplicate records. This indicates that no exact duplicate rows were identified during the data-quality check. Therefore, no duplicate records need to be removed at this stage.

```
df[["Age", "Purchase Amount (USD)", "Review Rating", "Previous Purchases"]].descr
```



	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3900.000000
mean	44.068462	59.764359	3.749949	25.351538
std	15.207589	23.685392	0.716223	14.447125
min	18.000000	20.000000	2.500000	1.000000
25%	31.000000	39.000000	3.100000	13.000000
50%	44.000000	60.000000	3.700000	25.000000
75%	57.000000	81.000000	4.400000	38.000000
max	70.000000	100.000000	5.000000	50.000000

```
numeric_cols = [
    "Age",
    "Purchase Amount (USD)",
    "Review Rating",
    "Previous Purchases"
]

for col in numeric_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1

    lower_limit = Q1 - 1.5 * IQR
    upper_limit = Q3 + 1.5 * IQR

    outliers = df[
        (df[col] < lower_limit) |
        (df[col] > upper_limit)
    ]

    print(f"{col}")
    print("Lower limit:", lower_limit)
    print("Upper limit:", upper_limit)
    print("Number of outliers:", len(outliers))
    print("-" * 40)
```

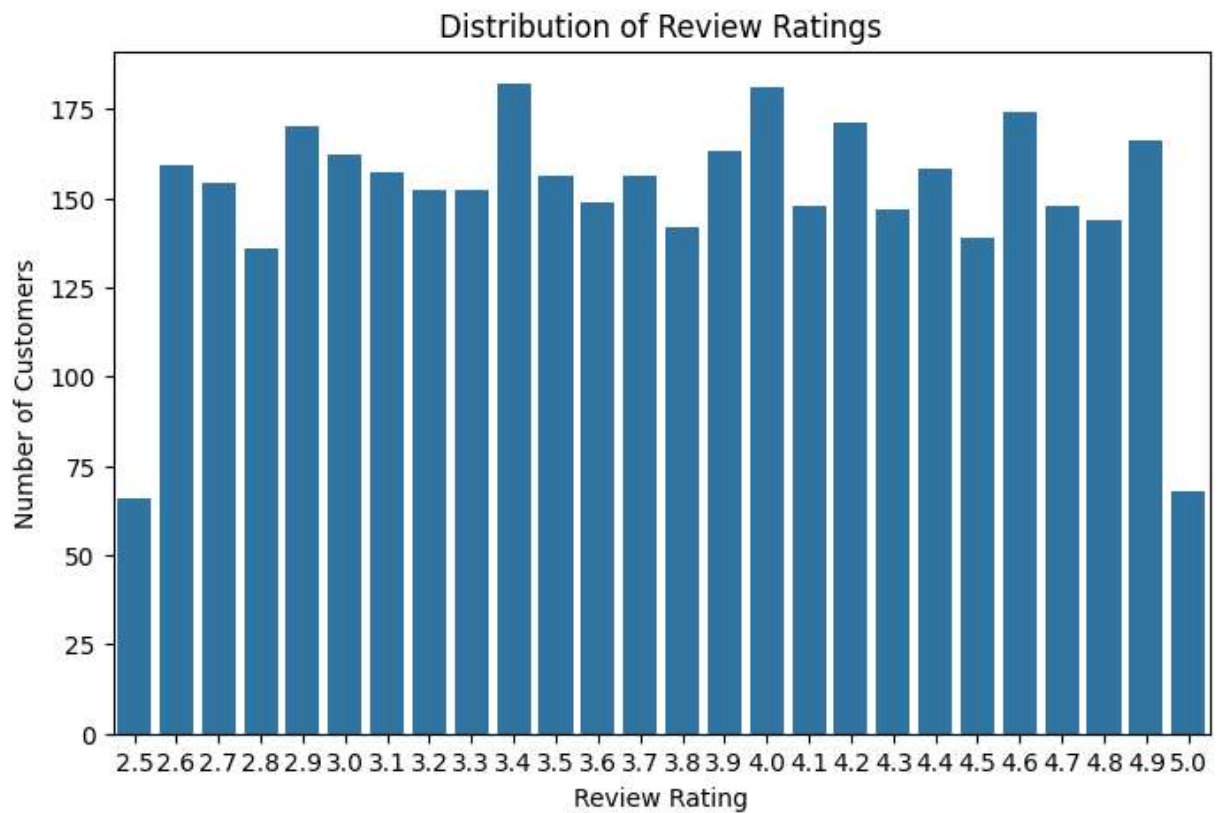
```
Age
Lower limit: -8.0
Upper limit: 96.0
Number of outliers: 0
-----
Purchase Amount (USD)
Lower limit: -24.0
Upper limit: 144.0
Number of outliers: 0
-----
Review Rating
Lower limit: 1.1499999999999997
Upper limit: 6.3500000000000005
Number of outliers: 0
-----
Previous Purchases
Lower limit: -24.5
Upper limit: 75.5
Number of outliers: 0
-----
```

```
plt.figure(figsize=(8,5))

sns.countplot(data=df, x="Review Rating")

plt.title("Distribution of Review Ratings")
plt.xlabel("Review Rating")
plt.ylabel("Number of Customers")

plt.show()
```



```
category_purchase = df.groupby("Category")["Purchase Amount (USD)"].mean().sort
print(category_purchase)
```

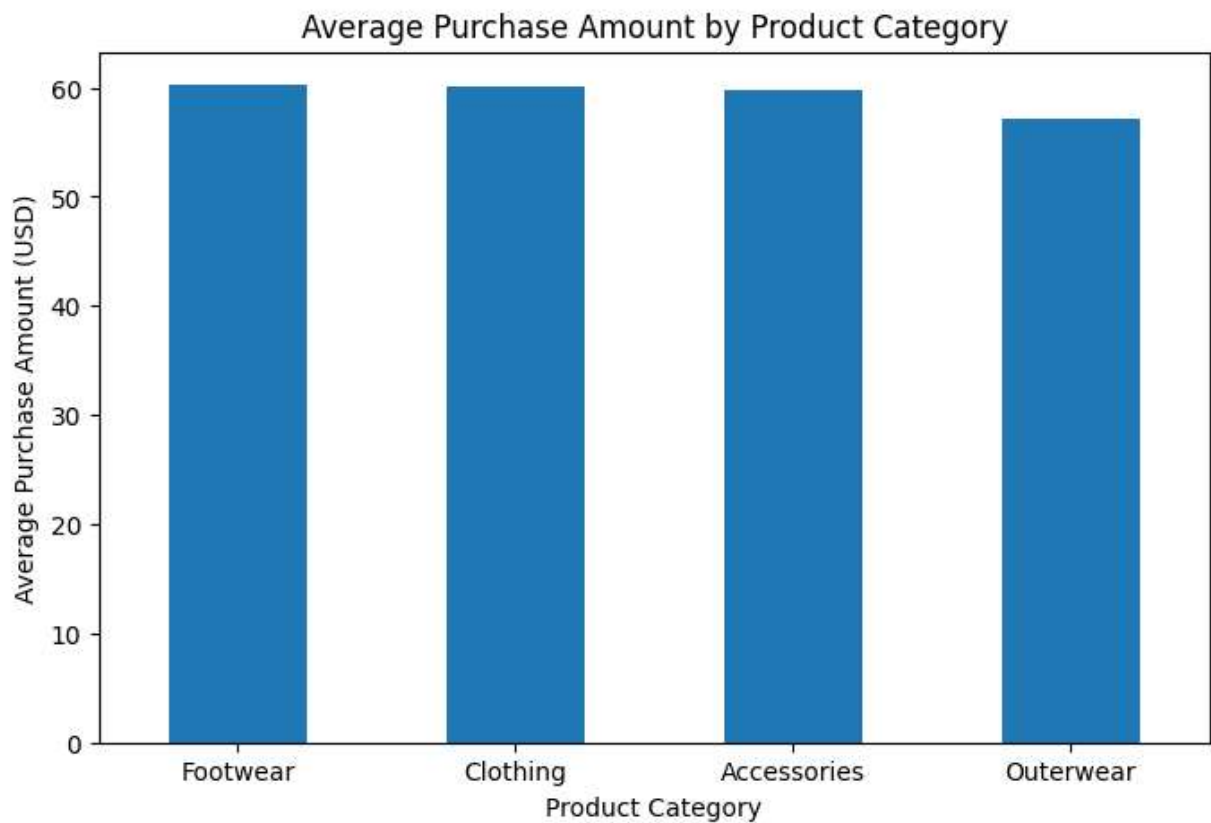
```
Category
Footwear      60.255426
Clothing      60.025331
Accessories   59.838710
Outerwear     57.172840
Name: Purchase Amount (USD), dtype: float64
```

```
plt.figure(figsize=(8,5))

category_purchase.plot(kind="bar")

plt.title("Average Purchase Amount by Product Category")
plt.xlabel("Product Category")
plt.ylabel("Average Purchase Amount (USD)")
plt.xticks(rotation=0)

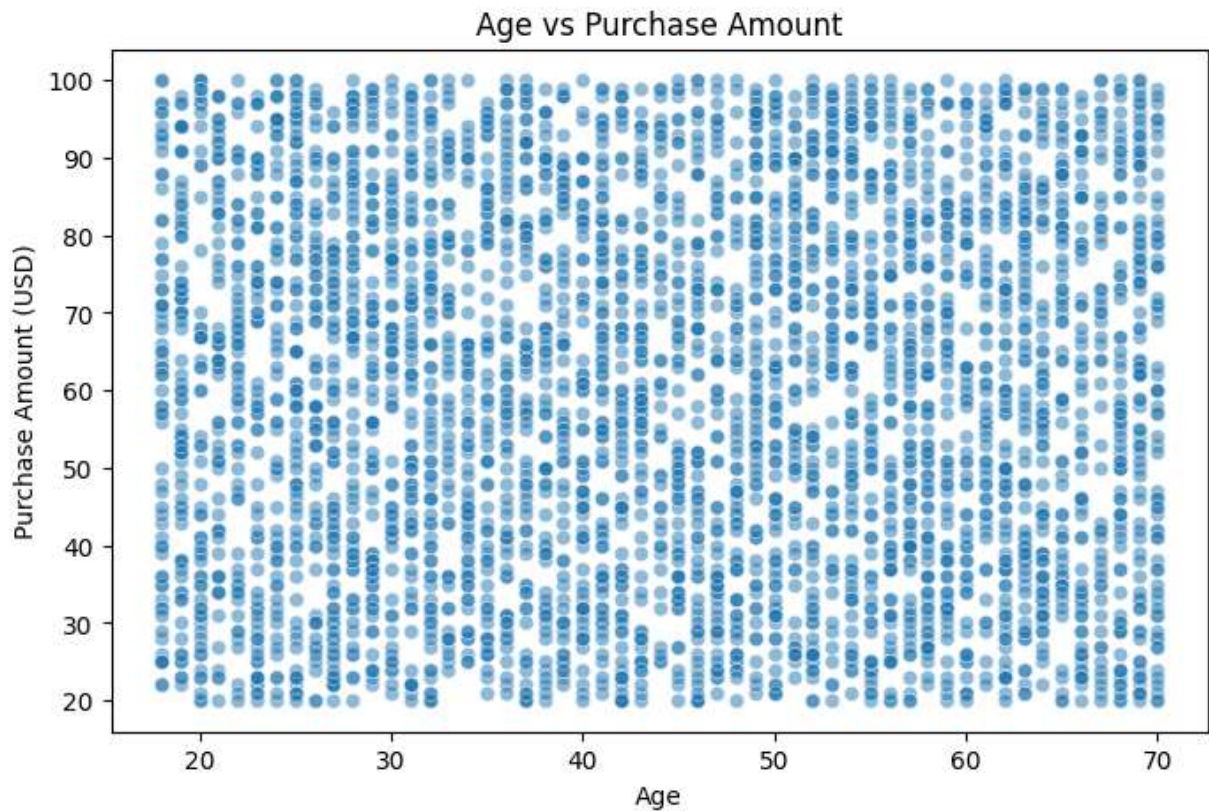
plt.show()
```



```
age_purchase_corr = df["Age"].corr(df["Purchase Amount (USD)"])  
print("Correlation between age and purchase amount:", age_purchase_corr)
```

Correlation between age and purchase amount: -0.01042364737868652

```
plt.figure(figsize=(8,5))  
  
sns.scatterplot(  
    data=df,  
    x="Age",  
    y="Purchase Amount (USD)",  
    alpha=0.5  
)  
  
plt.title("Age vs Purchase Amount")  
plt.xlabel("Age")  
plt.ylabel("Purchase Amount (USD)")  
  
plt.show()
```



```
category_rating = df.groupby("Category")["Review Rating"].agg(
    ["count", "mean", "median"]
).sort_values("mean", ascending=False)

print(category_rating)
```

	count	mean	median
Category			
Footwear	599	3.790651	3.8
Accessories	1240	3.768629	3.8
Outerwear	324	3.746914	3.8
Clothing	1737	3.723143	3.7

```
payment_counts = df["Payment Method"].value_counts()
print(payment_counts)
```

Payment Method	count
Credit Card	696
Venmo	653
Cash	648
PayPal	638
Debit Card	633
Bank Transfer	632

Name: count, dtype: int64

```
preference_counts = df["Preferred Payment Method"].value_counts()
print(preference_counts)
```

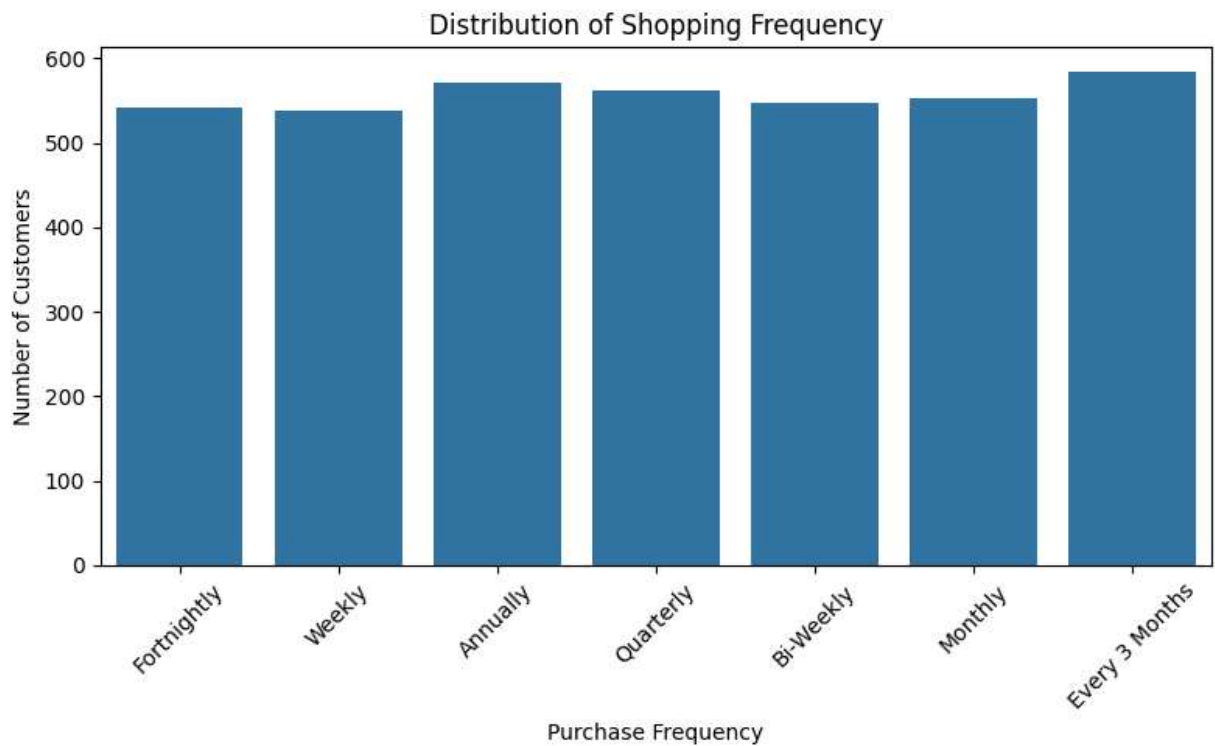
```
Preferred Payment Method
PayPal          677
Credit Card    671
Cash            670
Debit Card      636
Venmo           634
Bank Transfer   612
Name: count, dtype: int64
```

```
plt.figure(figsize=(8,5))

sns.countplot(
    data=df,
    x="Frequency of Purchases"
)

plt.title("Distribution of Shopping Frequency")
plt.xlabel("Purchase Frequency")
plt.ylabel("Number of Customers")

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```
print("missing values:")
print(df.isnull().sum())
print("\nduplicate values:")
print(df.duplicated().sum())
```

```
missing values:
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size             0
Color            0
Season           0
Review Rating    0
Subscription Status  0
Payment Method   0
Shipping Type    0
Discount Applied 0
Promo Code Used  0
Previous Purchases 0
Preferred Payment Method 0
```

```
Frequency of Purchases      0
dtype: int64
```

```
duplicate values:
0
```

```
# Checking outliers using IQR method

numerical_columns = ["Age", "Purchase Amount (USD)", "Review Rating", "Previous

for column in numerical_columns:
    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
    IQR = Q3 - Q1

    lower_limit = Q1 - 1.5 * IQR
    upper_limit = Q3 + 1.5 * IQR

    outliers = df[
        (df[column] < lower_limit) |
        (df[column] > upper_limit)
    ]

    print(column)
    print("Number of outliers:", len(outliers))
    print("-" * 40)
```

```
Age
Number of outliers: 0
-----
Purchase Amount (USD)
Number of outliers: 0
-----
Review Rating
Number of outliers: 0
-----
Previous Purchases
Number of outliers: 0
-----
```

```
from scipy.stats import pearsonr

data = df[["Age", "Purchase Amount (USD)"]].dropna()

correlation, p_value = pearsonr(
    data["Age"],
    data["Purchase Amount (USD)"]
)

print("Correlation:", correlation)
print("P-value:", p_value)
```

Correlation: -0.010423647378686533
P-value: 0.515197824122489

✓ Key Findings

After exploring the dataset, I found the following observations:

- The dataset contains information about customers, their purchases, product categories, ratings, payment methods and shopping frequency.
- The number of purchases is different for different product categories.
- The average purchase amount also varies across product categories.
- The review ratings are distributed across different rating values.
- Credit Card is the most commonly used payment method in the dataset.
- There are no missing values in the dataset.
- There are no duplicate records in the dataset.
- No outliers were found in Age, Purchase Amount, Review Rating and Previous Purchases using the IQR method.
- The correlation between Age and Purchase Amount is -0.0104, which shows that the relationship between these two variables is very weak.
- The p-value is 0.5152, which is greater than 0.05. Therefore, the analysis does not show a statistically significant linear relationship between Age and Purchase Amount.

Conclusion

Through this EDA, I was able to understand the main characteristics of the customers and their shopping behavior. I also checked the quality of the data by looking for missing values, duplicate records and outliers. The analysis of categories, purchase amounts, ratings and payment methods helped me understand the different patterns present in the dataset. These findings can be used for further analysis and visualization of customer shopping behavior.

```
# Correlation analysis between numerical variables

numeric_columns = [
    "Age",
    "Purchase Amount (USD)",
    "Review Rating",
    "Previous Purchases"
```

```

]

correlation_matrix = df[numeric_columns].corr()

print(correlation_matrix)

```

	Age	Purchase Amount (USD)	Review Rating	\
Age	1.000000	-0.010424	-0.021949	
Purchase Amount (USD)	-0.010424	1.000000	0.030776	
Review Rating	-0.021949	0.030776	1.000000	
Previous Purchases	0.040445	0.008063	0.004229	
	Previous Purchases			
Age		0.040445		
Purchase Amount (USD)		0.008063		
Review Rating		0.004229		
Previous Purchases		1.000000		

The correlation analysis shows that there is almost no linear relationship between Age and Purchase Amount, with a correlation of approximately -0.01. Purchase Amount and Review Rating also show a very weak positive relationship of about 0.03. Previous Purchases have very weak correlations with the other numerical variables. Overall, the relationships between these variables appear to be negligible in this dataset.

```

plt.figure(figsize=(8, 6))

sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Between Numerical Variables")
plt.tight_layout()
plt.show()

```

